

CSCE 753 COMPUTER VISION · MODULE 01 · LECTURE NOTES

The Inverse Problem

Why vision is hard in a way graphics is not.

MODULE OUTCOMES

- State the forward model and explain what inverting it loses.
- Explain ill-posedness with concrete ambiguities.
- Classify the four strategies for constraining the inversion.
- Explain why priors are unavoidable rather than a shortcut.
- Place every module in the course against that classification.

1 Forward and backward

RENDERING IS A FUNCTION; VISION INVERTS ONE THAT LOSES INFORMATION

CSCE 647's rendering equation maps a scene — geometry, materials, lights, camera parameters — to an image. Given the scene, the image is fully determined; the only difficulty is computing the integral, which is what that course was about.

Vision asks for the scene given the image, and the forward map is many-to-one. Enormous numbers of genuinely distinct scenes produce pixel-identical images — not approximately identical, but exactly so, as §2's scale argument shows.

So the inverse is not a function at all. There is no unique scene to recover, and any method that returns a single answer has supplied information from outside the image. Identifying what that information is, for any given method, is the most useful analytical habit in the subject.

That is the central fact of the course. Every technique here is a strategy for supplying the missing information (§3), and knowing which information a method assumes tells you precisely when it will fail — a stereo method fails on textureless walls because its missing information was correspondence; a learned depth model fails on an unusual room because its missing information was a prior over rooms.

Ambiguity	The confusion	What resolves it
Depth and scale	A small near object and a large far one project to identical pixels.	A second view with a known baseline, a known object size, or an inertial sensor (§2).
Albedo and illumination	A dark surface under bright light is indistinguishable from a bright surface under dim light.	A prior over materials and lighting, or control of the illumination. This is the intrinsic-image problem and it remains unsolved in general.

Ambiguity	The confusion	What resolves it
Shape and shading	A genuinely curved surface and a flat surface with a painted gradient produce the same intensities.	Motion (the gradient moves with the surface), or multiple known light directions (photometric stereo).
Specularity and texture	A specular highlight and a white painted patch look the same in one frame.	Viewpoint change — highlights move across the surface and paint does not , which is a two-frame test.
Occlusion	What lies behind an object is not attenuated or degraded; it is simply absent from the measurement.	Another viewpoint, or a learned prior about object completion. No amount of processing one image recovers it.
Bas-relief	A scene flattened along the view direction, with the lighting scaled correspondingly, produces an exactly identical image.	Nothing, from shading alone. The ambiguity is a proved theorem (Belhumeur, Kriegman and Yuille), which is why shape-from-shading cannot be made to work without additional constraints — and why decades of effort on it produced methods that need them.

2 In what sense ill-posed

THREE WAYS A PROBLEM CAN BE ILL-POSED

Existence — no scene explains the image. Rare in practice, and when it happens it almost always means the model is wrong (an unmodelled lens distortion, a rolling shutter) rather than that the image is impossible.

Uniqueness — many scenes explain the image equally well. **This is the dominant failure in vision** and the subject of §1's table.

Stability — a tiny change in the image produces a huge change in the recovered scene. **This is the condition that silently ruins working systems:** the answer is unique, the solver reports success, and the result is noise.

Stability is the condition-number problem arriving for a third time (CSCE 620 Module 01's robust predicates, CSCE 669 Module 05's convergence rate). **A near-degenerate configuration yields a unique answer with enormous error bars** — and Module 04's pure-rotation failure, where the essential matrix is unrecoverable because the baseline is zero, is exactly this. **The practical consequence is that you must report a conditioning estimate alongside every geometric estimate**, which working pipelines do and student code typically does not.

```
# A pinhole camera projects a 3D point to pixels by
#    $u = f * X / Z, \quad v = f * Y / Z$ 
#
# Scale the ENTIRE SCENE by k:
#    $u' = f * (kX) / (kZ) = f * X / Z = u$ 
#
# THE IMAGE IS BIT-IDENTICAL. Scene scale is unrecoverable
# from any number of images taken by a camera whose motion
# is itself unmeasured. Structure from motion returns
# geometry up to an unknown global scale, and that is not a
# limitation of the algorithm -- it is a property of
# perspective projection.
#
# TO FIX THE SCALE, something must be measured:
#   a known object size in the scene
#   a known camera baseline (a calibrated stereo rig)
#   an inertial measurement unit (visual-inertial odometry)
#   an active depth sensor
#
# This is why monocular SLAM drifts in scale, and why every
# phone AR system fuses the camera with the IMU rather than
# relying on vision alone.
```

Scale ambiguity is a theorem, not an engineering shortcoming. It is worth internalising early, because a great deal of time is lost trying to recover from a single camera something that provably is not there — and because the correct response (measure one thing) is cheap once the situation is understood.

3 The four strategies

- **More measurements.** More viewpoints, more light directions, more frames over time. **Modules 04 through 07** — and it is the **only strategy with provable guarantees**: two calibrated views in general position determine depth up to scale, and that is a theorem rather than an empirical finding.
- **Known structure.** Calibrated intrinsics, a planar scene, a rigid object, a printed target of known geometry. **Modules 02 and 03** — cheap when available, and the reason calibration is worth doing carefully rather than approximately.
- **Physical assumptions.** Brightness constancy, Lambertian reflectance, small inter-frame motion, smooth surfaces. **Modules 06 and 07** — powerful and nearly free, and **each assumption is precisely a failure mode**: optical flow fails on a specular surface because brightness constancy fails there, and knowing the assumption predicts the failure exactly.
- **Learned priors.** What scenes are plausible, learned from a large collection of them. **Modules 08 through 11** — **the only strategy that addresses semantic questions at all** ('is that a door?' is not a geometric question), and **the only one with no guarantees whatsoever**.
- **These are ordered by how much they can be trusted and inversely by how much they can do**, which is the trade this course negotiates from beginning to end. **A working system uses all four**, and the engineering judgement is about which part of the problem to give to which.

A PRIOR IS NOT CHEATING, AND IT IS NOT OPTIONAL

Because the inversion is under-determined, choosing an answer *requires* a preference over scenes. There is no neutral selection from an infinite set of equally consistent explanations.

Methods described as assumption-free have implicit priors — surface smoothness, Lambertian reflectance, general position, the absence of degenerate configurations. **These are no less assumptions for being unstated**, and being unstated makes them harder to test rather than weaker.

A learned prior is an explicit and measurable version of the same thing. Its genuine advantage over a hand-written assumption is not that it is more accurate but that **you can measure where it holds**, by evaluating on data drawn from a different distribution (Module 12).

Which is exactly why Module 12 is about distribution shift. A prior is as good as the match between its training distribution and the scene in front of you — and that match is a measurable quantity rather than something to be assumed, which makes it the central evaluation question for any learned vision component (CSCE 633 Module 12's discipline, applied to perception).

4 The shape of the course

WHERE THIS COURSE GOES

Modules 02 through 07 are geometric vision. Projective geometry and camera models, calibration, feature correspondence, two-view and multi-view geometry, bundle adjustment, dense stereo, and motion estimation. **Well-conditioned where its assumptions hold, provably correct, and the basis of every camera-tracking and mapping system that ships in a product.**

Modules 08 through 10 are learned vision. Convolutional features and transfer, detection and segmentation with their metrics, vision transformers, and self-supervised pretraining. **These answer the semantic questions geometry cannot touch** — identity, category, affordance — and offer no guarantees.

Module 11 is where the two meet. Neural radiance fields and Gaussian splatting reconstruct a scene by *optimising a renderer against photographs* — **CSCE 647's forward model used as a loss function, with CSCE 669's gradient methods as the optimiser.** It is the clearest single demonstration in this program of why the forward and inverse subjects belong in the same track.

Modules 12 and 13 are about what you can claim. Evaluation under distribution shift, benchmark pathologies, latency, calibration drift, and deployment. **The same closing discipline as every course in this program,** in the form it takes when the system's input is the physical world.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Szeliski — Computer Vision, chapter 1 (free PDF)	The framing of §1 and a survey of the whole subject, with a good history section.
Belhumeur, Kriegman & Yuille — The Bas-Relief Ambiguity (free)	The proved ambiguity from §1's table , and worth reading as an example of establishing that something is impossible.
Marr — Vision (1982), chapter 1	Where the inverse-problem framing of this module originates. Dated in its specifics and still the clearest statement of the problem. Library copy.
Hartley & Zisserman — Multiple View Geometry, chapter 1 (free sample)	The projective-geometry preview, which Module 02 develops properly.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Photograph two objects of very different sizes** so that they occupy identical image regions. Report the depths and sizes.

- 2 **Photograph a dark object in sunlight and a light object in shade** and measure the pixel intensities. Confirm the confusion.
- 3 **Render the same scene twice with all coordinates scaled by 2**, using your CSCE 647 renderer, and diff the images.
- 4 Confirm they are bit-identical, and explain why in one sentence.
- 5 **Photograph a specular surface from two viewpoints** and track where the highlight moves relative to the texture.
- 6 **Find a photograph where you cannot tell whether a surface is curved or painted**, and state what extra measurement would settle it.
- 7 **Take any five vision methods you have heard of** and classify each by which of the four strategies it uses.
- 8 **For each, name the assumption that constitutes its failure mode.**
- 9 **Find a paper claiming an assumption-free method** and identify its implicit prior.
- 10 **Write down, before the course proceeds, what you expect a camera to be able to measure about a scene.** Keep it and revisit after Module 07.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Why is the inverse of rendering not a function?
- 2 Name six single-image ambiguities and what resolves each.
- 3 Prove the scale ambiguity in two lines.
- 4 Why does every phone AR system use an inertial sensor?
- 5 Name Hadamard's three conditions and say which dominates in vision.
- 6 Which form of ill-posedness silently ruins working systems, and why?
- 7 List the four strategies in order of trustworthiness.
- 8 Why is a prior unavoidable rather than a shortcut?